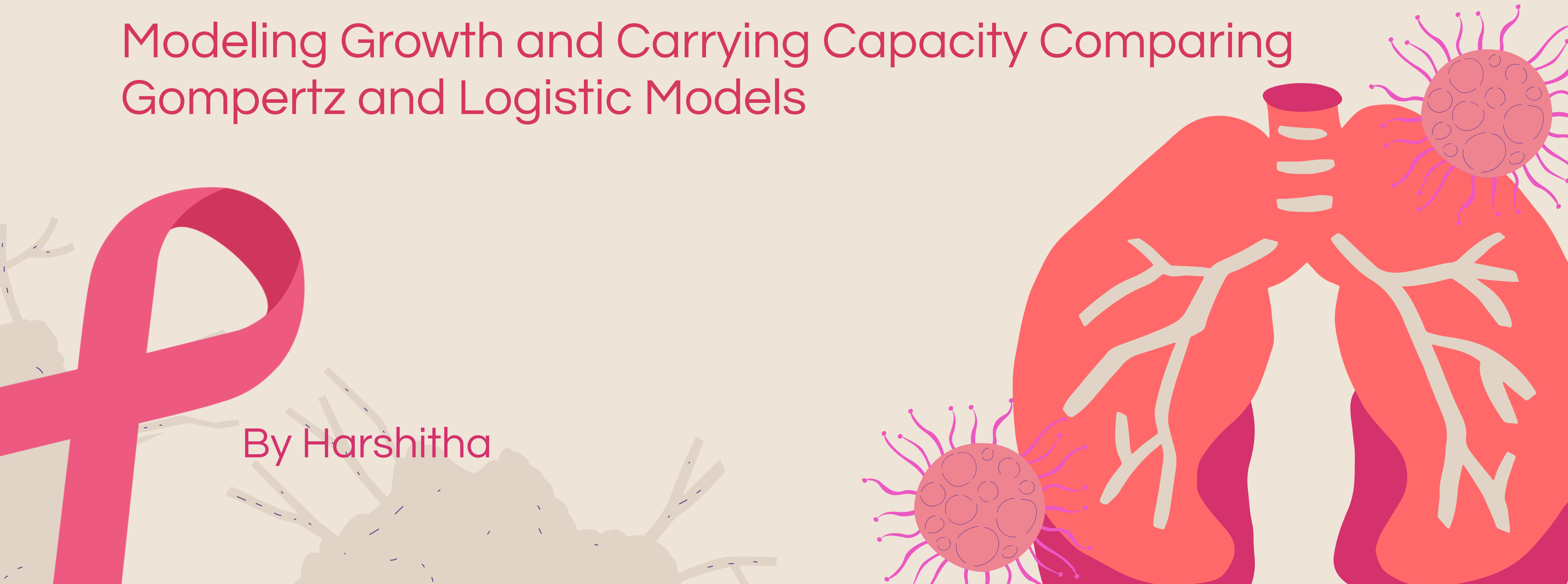


LEWIS LUNG CARCINOMA

Modeling Growth and Carrying Capacity Comparing
Gompertz and Logistic Models

By Harshitha



WHY THIS QUESTION MATTERS



1

1.5 million people die from from Lung Cancer every year

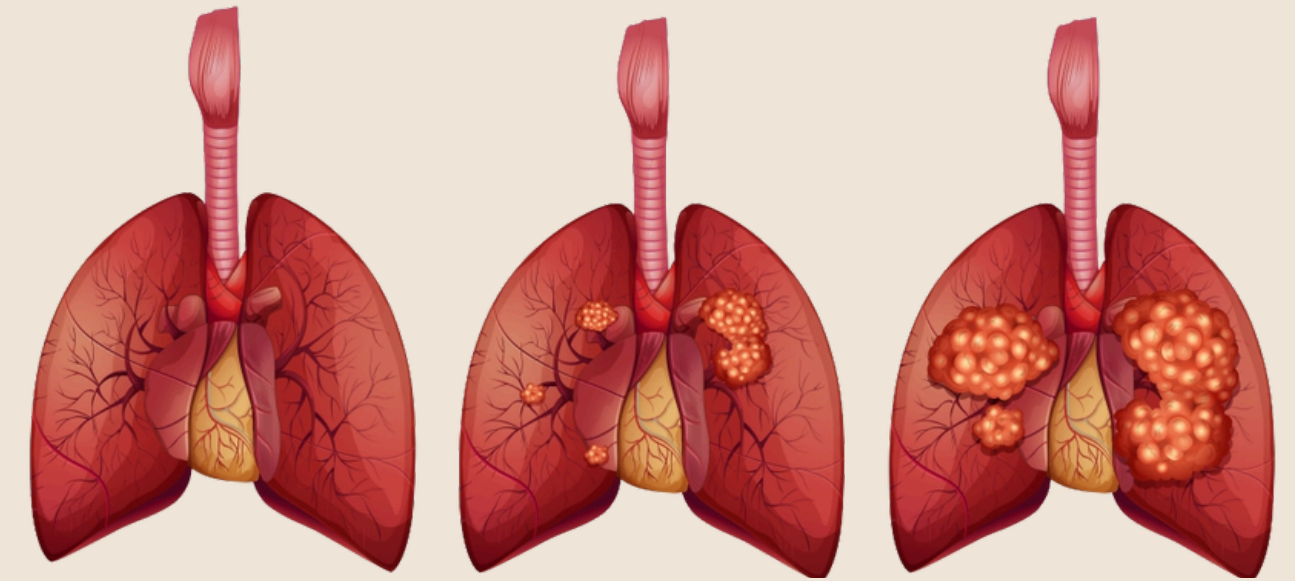
2

Doctors use differential equations to model tumor growth, diagnose the patient, and find pro

3

Translate complex biological behaviors into predictive mathematical models

WHY DOES LLC MATTER?



- Studying LLC more closely and mapping full growth has benefits
 - Improve treatment schedules
 - Design new medicine
 - Earlier diagnosis of tumors

WHAT IS LEWIS LUNG CARCINOMA (LLC)?



- It is a type of cancer in mice that is useful for studying lung cancer in humans.
- For studies, it is the leading preclinical model
- Mouse hosts must be euthanized (killed) before tumor growth slows down
 - Only early growth captured, ending never observed

WHY ARE THE MICE EUTHANIZED?

1

Tumor reaches a target size, the mice are sacrificed. This allows the researches to collect blood or tissue for cellular analysis.

2

To prevent unnecessary pain, distress, or spontaneous death, animal care guidelines are placed.

3

Fun Fact: They are killed by prolonged exposure to Carbon Dioxide or through anesthetic overdose



RESEARCH QUESTION

How accurately do the logistic and Gompertz models describe Lewis Lung Carcinoma growth and which is more biologically realistic?

1. Rearrange each differential equation to solve for K (carrying capacity)

2. Estimate K analytically using the data points

3. Regress using Desmos and cross-check with first estimation

4. Compare K values of logistic and Gompertz models to judge which is biologically plausible

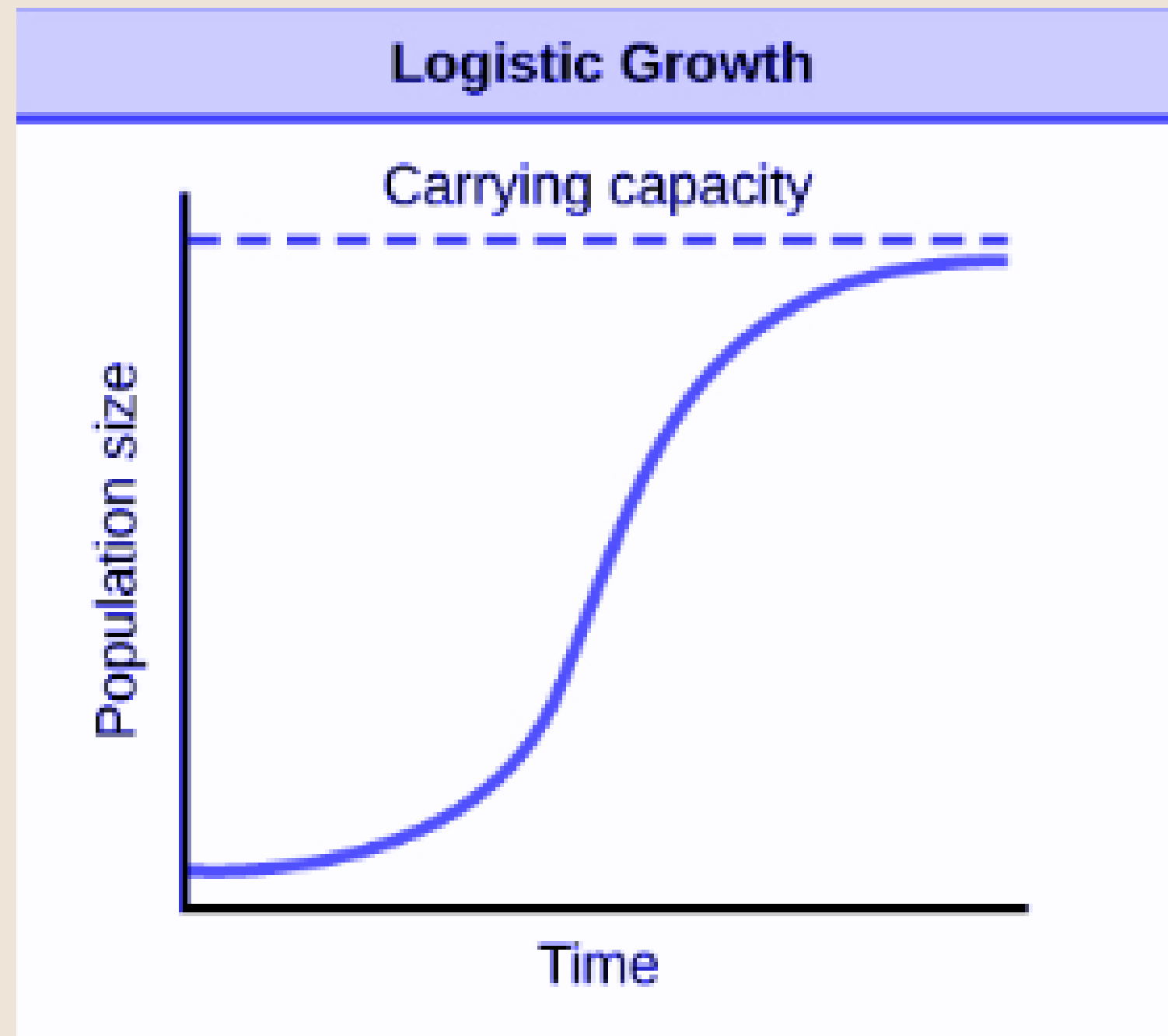
DEFINED VARIABLES

Symbol	Meaning	Units	Value Used
t	Time since injection	days	measured
V_0	Initial tumor volume	mm^3	1.0
r_L	Logistic growth-rate constant	day^{-1}	estimated
r_G	Gompertz growth-rate constant	day^{-1}	estimated
K	Carrying capacity	mm^3	estimated

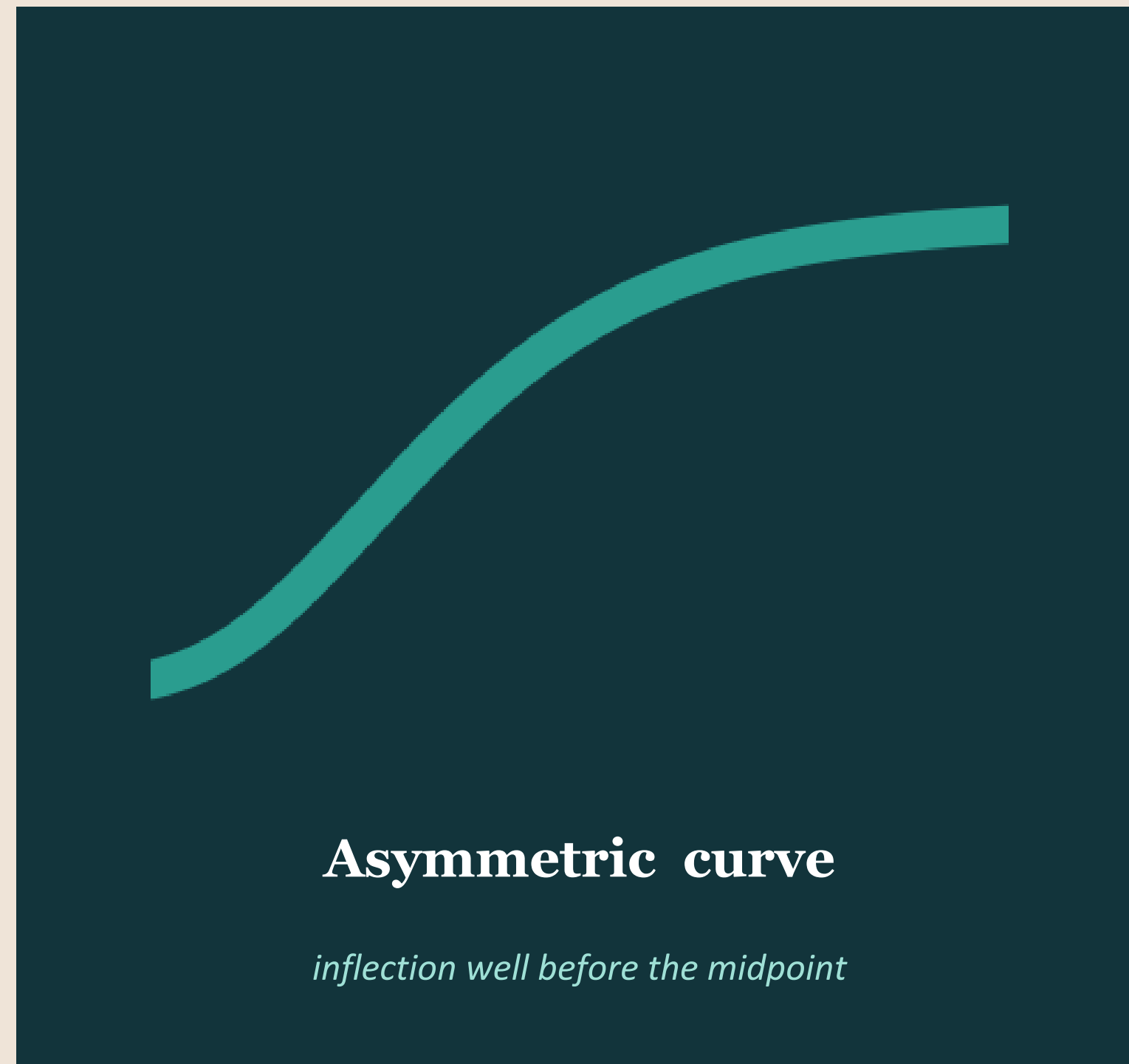
LOGISTIC VS GOMPERTZ

Logistic	Gompertz
Has an inflection exactly midway through tumor growth	Asymmetrical growth
Both flatten off with bounded growth, they do not grow forever	

LOGISTIC GRAPH EXAMPLE

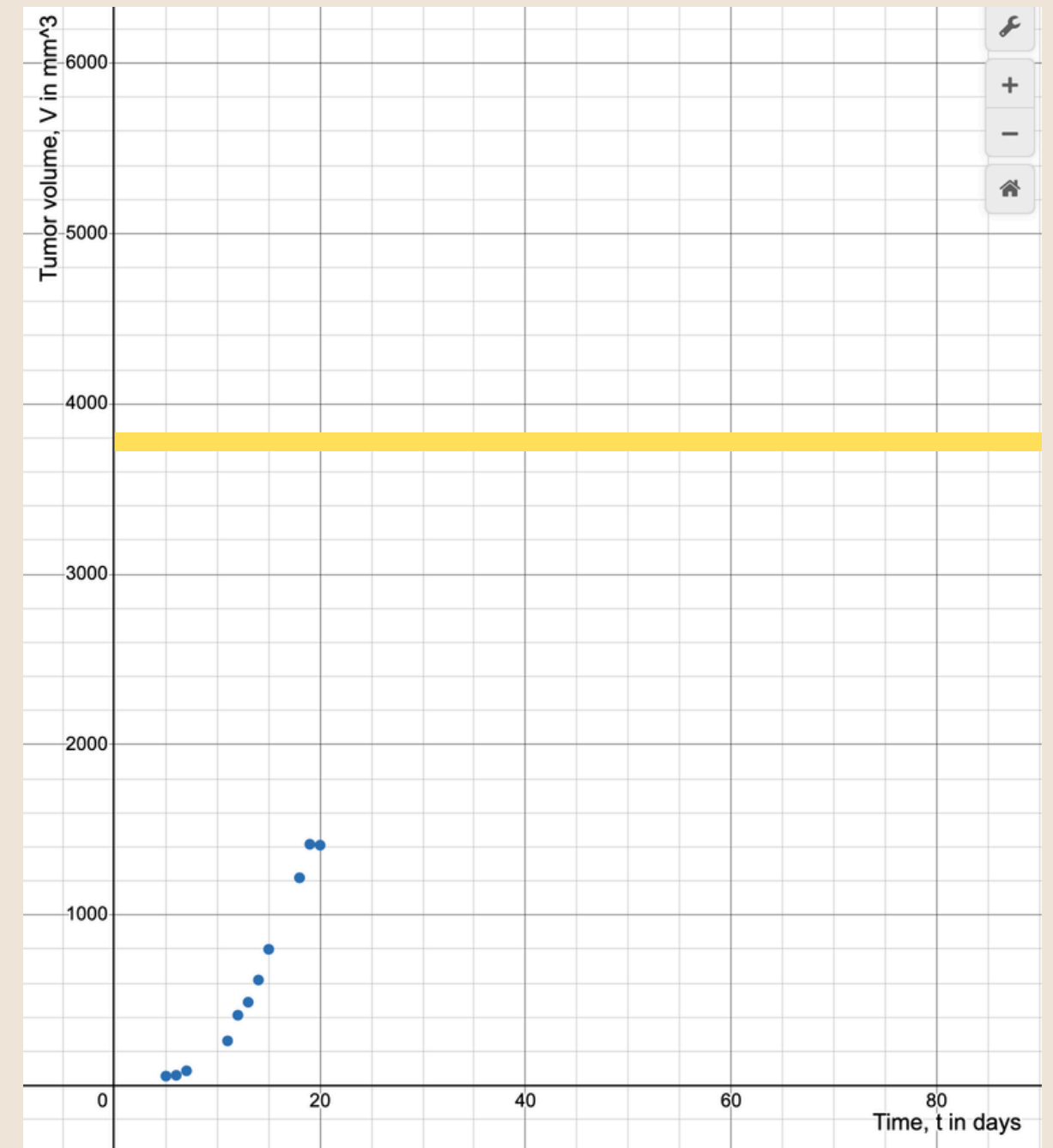


GOMPERTZ GRAPH EXAMPLE



WHY CARRYING CAPACITY (K)?

- All tumors slow down growth eventually
- With unfinished data, a biologically plausible K must sit well above the largest tumor measured
- By finding K from logistic and Gompertz models, we are able to tell which model is more biologically plausible for LLC



LOGISTIC MODEL

1. Separate variables by moving V to one side and t to the other
2. Use partial fractions to rewrite the equation
3. Integrate both sides
4. Solve for V(t)

$$\frac{dV}{dt} = r_L V \left(1 - \frac{V}{K} \right)$$



$$V(t) = \frac{K}{1 + B e^{-r_L t}}$$



$$K = \frac{V(1 - e^{-r_L t})}{1 - V e^{-r_L t}}$$

This allows us to plug in data to solve for K and r systematically

REARRANGING GOMPERTZ

Substitute

Let $u = \ln(K/V)$. The equation becomes linear in u .

Reduce

It collapses to $du/dt = -r \cdot u$ (easy to separate).

Integrate

Solve for u , then back-substitute the log.

Solve for $V(t)$

Exponentiate twice to free the volume.

$$\frac{dV}{dt} = r_G V \ln \left(\frac{K}{V} \right)$$

$$V(t) = K e^{-\ln(K)} e^{-r_G t}$$

$$K = e^{\left(\frac{\ln(V)}{1 - e^{-r_G t}} \right)}$$

This allows us to plug in data to solve for K and r systematically

WHY GOMPERTZ - REAL WORLD

Main Idea:

- Gompertz accurately reflects how real tumors actually grow over time

Tumors DO NOT grow at a constant exponential rate forever.

Instead, they follow these three stages:

1. Slow initial Growth

- a. At first, only a few cancer cells exist, so growth is relatively slow.

2. Rapid middle growth

- a. Once the tumor establishes itself and cells divide quickly, growth accelerates

3. Slowing growth later

- a. As the tumor becomes larger, growth slows because of limited nutrients, oxygen, blood supply, and space.
- b. Cells also compete with one another